

Rasmus Frigaard Lemvig - Curriculum Vitae

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Homepage: <https://rasmusfl.github.io/>

Education

University of Copenhagen

PhD student. Principal supervisor: Martin Bladt. Co-supervisor: Mogens Steffensen (*May 2025 - present*)

Working thesis title: *Multi-state models: modern methods, machine learning and meditations* ((MM)³)

University of Copenhagen

Master of Science in Actuarial Mathematics (*Sep 2023 - Jan 2025*)

Thesis: *Machine learning methods for survival and multi-state models*

University of Copenhagen

Bachelor of Science in Actuarial Mathematics (*Sep 2021 - Jun 2023*)

Two-part thesis on reserving under stress scenarios in life insurance and insurance pricing using GLMs and machine learning.

University of Copenhagen

Bachelor of Science in Mathematics (*Sep 2018 - Jun 2021*)

Thesis: *Cubic and Quartic Reciprocity*

Experience

University of Copenhagen - Department of Mathematical Sciences

PhD student (*May 2025 - present*). PhD student in the section of Insurance and Economics (IE). Interests include the mathematics and statistics of life insurance, in particular multi-state modelling, and their intersection with machine learning. Other interests are mathematical aspects of risk management and finance.

Scientific assistant (*Feb 2025 - May 2025*).

Teaching assistant (*Sep 2020 - present*) 24 times in total, spanning 20 different courses. Noteworthy courses include:

- Statistics for Insurance
- Quantitative Risk Management
- Topics in Life Insurance
- Topics in Probability
- Basic Life Insurance Mathematics
- Statistical Methods
- Brownian Motion
- Continuous Time Finance
- Advanced Probability Theory 2
- Stochastic Processes in Non-Life Insurance
- Probability Theory 2

- Probability Theory

My complete teaching history can be found here: <https://rasmusfl.github.io/work.html>.

Manuscripts and projects

1. *Consistency of Honest Decision Trees and Random Forests*
Martin Bladt and Rasmus Frigaard Lemvig
 2. *Local estimation of transition rates of jump processes through discretization*
Martin Bladt and Rasmus Frigaard Lemvig
 3. *JumpForests: A random forest R package for jump processes*
Martin Bladt and Rasmus Frigaard Lemvig
 4. *Lecture notes for the course Quantitative Risk Management (QRM)*
Rasmus Frigaard Lemvig
 5. *Lecture notes for the course Stochastic Processes in Non-Life Insurance (SkadeStok) 2023/2024*
Rasmus Frigaard Lemvig
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Programming proficiencies

Experience with programming in **R**, **Julia**, **Python/Cython** and **C++**. Developed an R package for fitting random forests for jump processes using **RCpp**.

Volunteer work

Danish Youth Association of Science (UNF)

Volunteer (*Jan 2016 - present*). Some highlights include:

- **Cofounder of, coordinator and teacher at UNF Science Weekend** (2019 - 2023), a biannual weekend camp for elementary school students with 85 participants and 30 volunteers.
- **Teacher at Mathematics Camp** from 2021 to 2025 and at Machine Learning Camp 2024.
- **Teacher at Week in Mathematical Modelling 2025**, a camp with focus on differential equations and their applications. Teacher and coordinator in 2026 also.
- **Teacher** in several workshops. Topics: Linear algebra, calculus, number theory, graph theory, sums and sequences, differential equations.
- **Coordinator of and teacher in** the project Evenings of Natural Science Education where we taught young Ukrainian refugees.

A detailed description of my work for UNF can be found here: <https://rasmusfl.github.io/volunteer.html>.

University of Copenhagen - Department of Mathematical Sciences

Student representative in the Board of Education (*June 2022 - January 2025*)